

NAME AND SURNAME

M.S. Thesis

2016

THESIS TITLE

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IŞIK UNIVERSITY

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NAME AND SURNAME

B.S., Computer Engineering, IŞIK UNIVERSITY, 2009

B.S., Mathematics Engineering, IŞIK UNIVERSITY, 2009

Submitted to the Graduate School of Science and Engineering
in partial fulfillment of the requirements for the degree of
Master of Science
in
Computer Engineering

IŞIK UNIVERSITY

2016

IŞIK UNIVERSITY
GRADUATE SCHOOL OF SCIENCE AND ENGINEERING

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APPROVAL DATE: / /

THESIS TITLE

Abstract

Deepfake technology has rapidly evolved due to advancements in generative AI models such as GANs, diffusion models, and neural voice synthesis systems. While these technologies enable realistic synthetic media generation, they also introduce significant cybersecurity threats including identity fraud, misinformation, and biometric spoofing. Existing detection approaches often rely on single-modality analysis (visual or audio), which limits robustness against modern high-quality deepfakes. This research investigates a multimodal deepfake detection framework that integrates visual, audio, and temporal features to improve detection accuracy and generalization. The proposed system leverages convolutional neural networks, transformer-based encoders, and attention-based fusion mechanisms to identify inconsistencies across modalities. Benchmark datasets such as FaceForensics++, DFDC, Celeb-DF, and ASVspoof are considered for evaluation. The expected outcome is a more robust and generalizable detection system suitable for real-world cybersecurity applications. [1]-[2]

Keywords: Deepfake detection, multimodal AI, cybersecurity, GAN forensics, audio-visual analysis, temporal modeling

TEZ BAŐLIĐI

Özet

Anahtar kelimeler:

Acknowledgements

This study was supported by The Scientific and Technological Research Council of Turkey (TÜBİTAK) Grant No:

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List of Abbreviations

GAN	G enerative A dversarial N etwork
CNN	C onvolutional N eural
LSTM	L ong S hort- T erm M emory
MFCC	M el F requency C epstral C oefficients
DFDC	D eep F ake D etection C hallenge
ASV	A utomatic S peaker V erification

Chapter 1

Introduction

Deepfakes are AI-generated or manipulated multimedia content designed to mimic real human identities. With the rapid development of deep generative models, deepfakes have become highly realistic and difficult to detect. These technologies pose serious risks in cybersecurity, including impersonation attacks, fake news dissemination, and biometric authentication bypass [3] [4].

Traditional detection systems rely primarily on visual artifacts; however, modern deepfakes minimize such inconsistencies. This has led to increased interest in multimodal detection systems that analyze both audio and visual inconsistencies along with temporal behavior [1] [5]. The objective of this research is to develop a robust multimodal framework capable of detecting deepfakes under real-world conditions.

Chapter 2

Literature Survey

2.1 Visual Detection Methods

Early research focused on visual inconsistencies in manipulated videos. Works such as FaceForensics++ by Andreas Rossler introduced benchmark datasets and CNN-based detection models [1]. Architectures like XceptionNet and EfficientNet were widely used. These methods detect:

- texture artifacts
- blending errors
- lighting inconsistencies

However, they fail under compression and unseen manipulations [3].

2.2 Audio-Based Detection Methods

Audio deepfake detection uses MFCC features, spectrograms, and transformer models. ASVspoof datasets became standard benchmarks [2]. Despite good performance on known attacks, modern neural voice synthesis reduces detectable artifacts significantly.

2.3 Temporal and Multimodal Methods

Temporal models analyze motion consistency across frames, detecting blinking patterns and lip-sync mismatches [5]. Multimodal approaches combine audio-visual features using fusion networks. These systems outperform single-modality models but still face:

- high-computational cost
- poor explainability
- limited real-time use

2.4 Research Gap

Existing systems fail to generalize across datasets and rely heavily on modality-specific artifacts. This motivates multimodal cross-consistency analysis to improve robustness.

2.5 Relation to Research Question

This directly supports the research question:

How can multimodal deepfake detection improve robustness compared to single-modality approaches?

Chapter 3

Approach

3.1 Overall Approach

The proposed system integrates visual, audio, and temporal analysis to detect inconsistencies across modalities. Unlike traditional systems, it focuses on cross-modal verification rather than isolated detection.

3.2 System Architecture

The system includes:

- **Visual Module:** CNN-based feature extraction (Xception/EfficientNet)
- **Audio Module:** Transformer-based speech analysis
- **Temporal Module:** Sequential frame consistency analysis
- **Fusion Module:** Attention-based integration

Final classification outputs real/fake probability.

3.3 Dataset

- FaceForensics++ [1]
- Celeb-DF
- DFDC
- ASVspoof [2]

Preprocessing includes face extraction, audio segmentation, and normalization.

3.4 Expected Contribution

- Improved multimodal detection accuracy
- Better generalization across datasets
- Enhanced robustness against modern deepfakes
- Potential deployment in cybersecurity systems

Chapter 4

Results and Discussion

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Chapter 5

Conclusion

comming soon...

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